The documentation and process conversion measures necessary to comply with this document shall be completed by 23 July 2004.

INCH-POUND

MIL-PRF-19500/349F
23 April 2005
SUPERSEDING
MIL-PRF-19500/349E
28 August 2000

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPES 2N3506, 2N3506A, 2N3506L, 2N3506AL, 2N3506U4, 2N3507, 2N3507L, 2N3507A, 2N3507AL, 2N3507U4, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to T0-39) and figure 2 (U4).

* 1.3 Maximum ratings unless otherwise specified T_A = +25°C. (1)

Types	P_T T_A = +25°C (1)	P_T T_C = +25°C (1)	$R_{\theta JA}$	$R_{\theta JC}$	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_J and T_{STG}
	W	W	°C/W	°C/W	V dc	V dc	V dc	A dc	°C
2N3506, 2N3506A,	1.0	N/A	175	18	60	40	5.0	3.0	
2N3506L, 2N3506AL	1.0	N/A	175	18	60	40	5.0	3.0	
2N3506U4	N/A	5.0	N/A	7	60	40	5.0	3.0	-65 to
2N3507, 2N3507A,	1.0	N/A	175	18	80	50	5.0	3.0	+200
2N3507L, 2N3507AL	1.0	N/A	175	18	80	50	5.0	3.0	
2N3507U4	N/A	5.0	N/A	7	80	50	5.0	3.0	

(1) For derating see figures 3 and 4.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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1.4 Primary electrical characteristics. (1)

	h_{FE2} (1)		h_{FE4} (1)		$V_{CE(sat)2}$	$\|h_{fe}\|$	C_{obo}	t_{on}	t_{off}
Limits	V_{CE} = 2.0 V dc I_C = 1.5 A dc		V_{CE} = 5.0 V dc I_C = 3.0 A dc		I_C = 1.5 A dc I_B = 150 mA dc	f = 20 Mhz V_{CE} = 5 V dc I_C = 100 mA dc	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	I_C = 1.5 A dc I_B = 150 mA dc	
	2N3506, 2N3506A, 2N3506L, 2N3506AL, 2N3506U4	2N3507, 2N3507A, 2N3507L, 2N3507AL, 2N3507U4	2N3506, 2N3506A, 2N3506L, 2N3506AL, 2N3506U4	2N3507, 2N3507A, 2N3507L, 2N3507AL, 2N3507U4					
					V dc		pF	ns	ns
Min	40	30	25	20		3			
Max	200	150			1.0	15	40	45	90

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

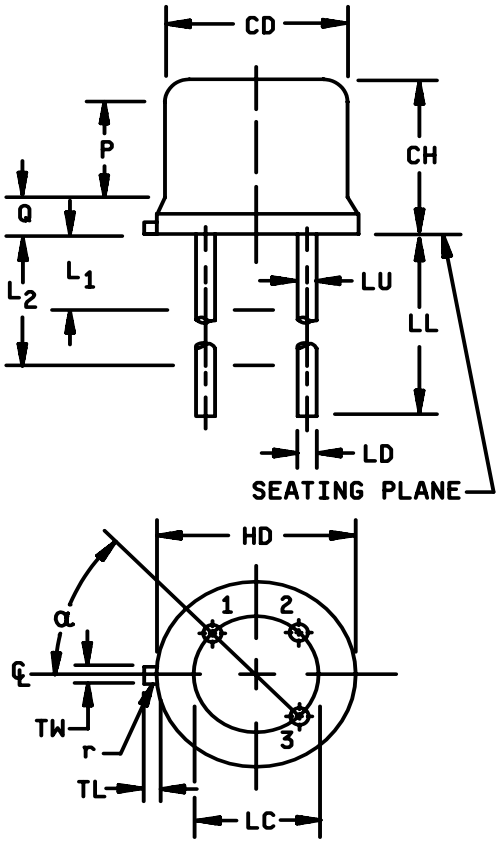
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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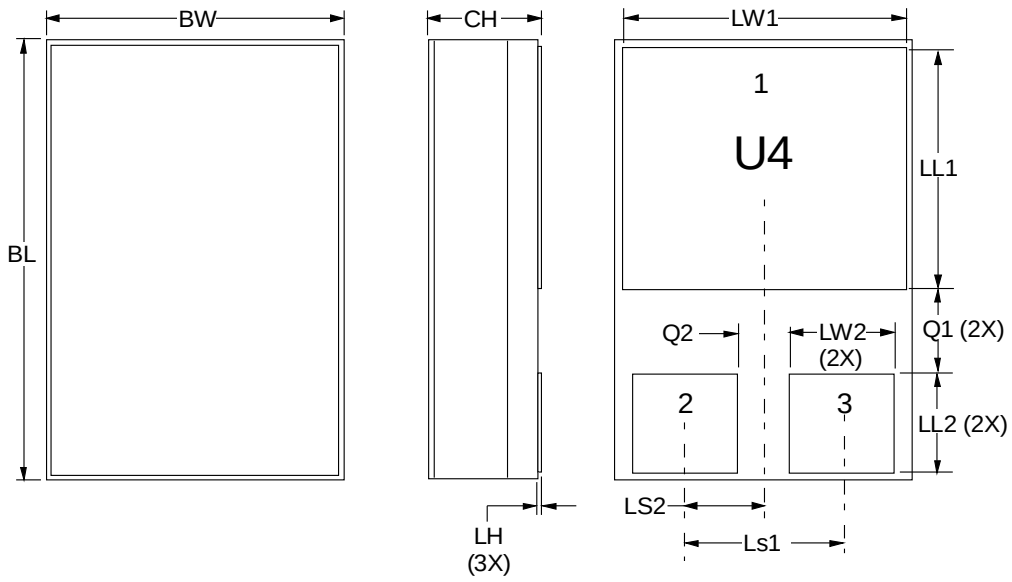
Dimensions					
Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CD	0.305	0.335	7.75	8.51	
CH	0.240	0.260	6.10	6.60	
HD	0.335	0.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	0.016	0.021	0.41	0.53	7, 8
LL	See notes				7, 8, 11,12
LU	0.016	0.019	0.41	0.48	7, 8
L1		0.050		1.27	7, 8
L2	0.250		6.35		7, 8
P	0.100		2.54		5
Q		0.050		1.27	4
r		0.010		0.25	10
TL	0.029	0.045	0.74	1.14	3
TW	0.028	0.034	0.71	0.86	2
α	45/ TP		45/ TP		6



NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. For 2N3506L, 2N3507L, 2N3506AL, and 2N3507AL, dimension LL shall be 1.5 inches (38.1mm) minimum and 1.75 inches (44.4 mm) maximum.
12. For 2N3506, 2N3506A, 2N3507, and 2N3507A dimension LL shall be .5 inches (12.7mm) minimum and 0.75 inches (19.0 mm) maximum.
13. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
14. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions (similar to TO-39).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.215	.225	5.46	5.72
BW	.145	.155	3.68	3.94
CH	0.049	0.075	1.24	1.91
LH		0.005		0.125
LW1	.135	.145	3.43	3.68
LW2	.047	.057	1.19	1.45
LL1	0.115	0.125	2.91	3.17
LL2	0.045	0.055	1.14	1.64
LS1	0.070	0.095	1.48	2.41
LS2	0.038	0.048	0.825	1.21
Q1	.045	.070	1.14	1.78
Q2	0.025	0.035	0.635	0.88
1	Collector			
2	Base			
3	Emitter			

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 2. Physical dimensions, surface mount (2N3506U4, 2N3507U4) version (U4).

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3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

$R_{\theta JA}$	Thermal resistance junction to ambient.
$R_{\theta JC}$	Thermal resistance junction to case.
TRB	Technical review board.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (similar to T0-39) and figure 2 (U4).

* 3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and, tables I, and II).

* 4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, 4.2.1, and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance, method 3131 of MIL-STD-750, (see 4.3.2)	Thermal impedance, method 3131 of MIL-STD-750, (see 4.3.2)
7	hermetic seal (optional)	Optional
9	I_{CEX1}, h_{FE2}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CEX1}; h_{FE2}; ΔI_{CEX1} = 100 percent of initial value or 200 nA dc, whichever is greater; Δh_{FE2} = 15 percent of initial value	I_{CEX1} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX1} = 100 percent of initial value or 200 nA dc, whichever is greater; Δh_{FE2} = 15 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CEX1} = 100 percent of initial value or 200 nA dc, whichever is greater; Δh_{FE2} = 15 percent of initial value.
14	Required	Required

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) maybe used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

* 4.3.2 Thermal impedance. See figures 5, 6, and 7. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The thermal impedance limit shall comply with the thermal impedance graph in figures 5, 6, and 7 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3131 of MIL-STD-750. See Group E subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. See 4.4.2.2 and table VIb, of MIL-PRF-19500, for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	2037	Test condition A
B4	1037	V_{CB} = 10 V dc
B5	1027	V_{CB} = 10 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with table VIa of MIL-PRF-19500, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjusted T_A or P_D to achieve a T_J = +225°C minimum.
B6	3131	See 4.5.2

* 4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failures mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum power dissipation P_D = 75 percent of P_T maximum rated as defined in 1.3 herein. n= 45 devices, c = 0. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. Samples shall be selected from each wafer from a die fabrication line that the manufacturer chooses to certify for the die design used in products manufactured to this specification.

b. Must be chosen from an inspection lot that has been submitted to and passed group A conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B3 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

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4.4.2.4 Product acceptable for delivery. The product shall be acceptable for delivery only after the successful completion of all group B testing and shall be comprised of die meeting the following requirements:

a. Manufactured on the same certified die fabrication line as the sample selected for 4.4.2.3.

b. The die lot was manufactured (using start or completion dates, as the manufacturer selects to consistently use) within three months following the wafer lot used for sample selection in 4.4.2.3. Group B shall not be required more than once for any single wafer lot.

4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3).
C6	1026	Steady-state life: 1,000 hours, V_{CB} = 10 dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

* 4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	3131	$R_{\theta JA}$ (see 1.3).
C6		Not applicable

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.3.4 Product acceptable for delivery. The product which may be accepted for delivery based a successful completion of group C testing shall be the product of the particular package type with lot inspection identification codes of the 52 successive weeks beginning with the inspection lot identification code of the successful group C sample. (I.E. the inspection lot identification codes accepted for delivery based on successful group C testing will include those sealed during the approximate 10 weeks during which the sample was being screened and tested plus those sealed in the 42 weeks following successful completion of group C testing making a total of 52 successive calendar weeks.) A manufacturer shall not accept product for delivery until after successful completion of the group C testing on product of the applicable package type with an inspection lot identification code less than 51 weeks prior to the inspection identification code of the product being considered for acceptance.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 3/						
Visual and mechanical 4/ examination	2071	n = 45 devices, c = 0				
Solderability 4/ 5/	2026	n = 15 leads, c = 0				
Resistance to solvents 4/ 5/ 6/	1022	n = 15 devices, c = 0				
Temp Cycling 4/ 5/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic Seal 5/	1071	n = 22 devices, c = 0				
Fine leak						
Gross leak						
Electrical measurements 5/		Table I, subgroup 2				
Bond strength 4/ 5/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = 300°C at t = 2 hrs n = 11 wires, c = 0				
Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
Subgroup 2						
Thermal impedance 7/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Breakdown voltage collector to base	3001	Bias condition D, I_C = 100 µA dc	$V_{(BR)CBO}$			
2N3506				60		V dc
2N3507				80		V dc
Breakdown voltage emitter to base	3026	Bias condition D, I_E = 10 µA dc	$V_{(BR)EBO}$	5		V dc
Breakdown voltage collector to emitter	3011	Bias condition D, I_C = 10 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3506				40		V dc
2N3507				50		V dc
Collector to emitter cutoff current	3041	Bias condition A, V_{EB} = 4 V dc	I_{CEX1}		1	µA dc
2N3506		V_{CE} = 40 V dc				
2N3507		V_{CE} = 60 V dc				
Forward-current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE1}			
2N3506				50	250	
2N3507				35	175	

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 1.5 A dc, pulsed (see 4.5.1)	h_{FE2}			
2N3506				40	200	
2N3507				30	150	
Forward-current transfer ratio	3076	V_{CE} = 3 V dc, I_C = 2.5 A dc, pulsed (see 4.5.1)	h_{FE3}			
2N3506				30		
2N3507				25		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 3.0 A dc, pulsed (see 4.5.1)	h_{FE4}			
2N3506				25		
2N3507				20		
Saturation voltage and resistance	3071	I_C = 500 mA dc, I_B = 50 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.5	V dc
Saturation voltage and resistance	3071	I_C = 1.5 A dc, I_B = 150 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.0	V dc
Saturation voltage and resistance	3071	I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)3}$		1.5	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 500 mA dc, I_B = 50 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.0	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 1.5 A dc, I_B = 150 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$	0.8	1.3	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)3}$		2.0	V dc
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{EB} = 4 V dc.	I_{CEX2}		25	µA dc
2N3506		V_{CE} = 40 V dc				
2N3507		V_{CE} = 60 V dc				
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE5}			
2N3506, 2N3506L		V_{CE} = 1.0 V dc		25		
2N3507, 2N3507L		V_{CE} = 1.0 V dc		17		
2N3506A, 2N3506AL		V_{CE} = 2.0 V dc		25		
2N3507A, 2N3507AL		V_{CE} = 2.0 V dc		17		

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Small-signal short- circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc, I_C = 100 mA dc, f = 20 MHz	$\|h_{fe}\|$	3	15	
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		40	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 3 V dc, I_C = 0, 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		300	pF
Pulse response:						
Delay time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = 150 mA dc, (see figure 8)	t_d		15	ns
Rise time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = 150 mA dc, (see figure 8)	t_r		30	ns
Storage time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = I_{B2} = 150 mA dc, (see figure 9)	t_S		55	ns
Fall time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = I_{B2} = 150 mA dc, (see figure 9)	t_f		35	ns

1/ For sampling plan see MIL-PRF-19500.
2/ Electrical characteristics for "A", "AL", and "L" suffix devices are identical to non-suffix devices unless otherwise noted.
3/ For resubmission of failed subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
4/ Separate samples may be used.
5/ Not required for JANS devices.
6/ ot required for laser marked devices.
7/ This test is required for the following end-point measurements only (not intended for screen 13). JANS - group B, subgroup 3 and 4. JANTX and JANTXV – group B, subgroup 2 and 3; group C, subgroup 6; group E, subgroup 1.

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* TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of +100°C.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			
Thermal impedance curves		Each supplier shall submit their qualification lot average and design maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report.	Sample size N/A
Subgroup 5			
Not applicable			
Subgroup 6			3 devices
ESD (electrocstatic discharge)	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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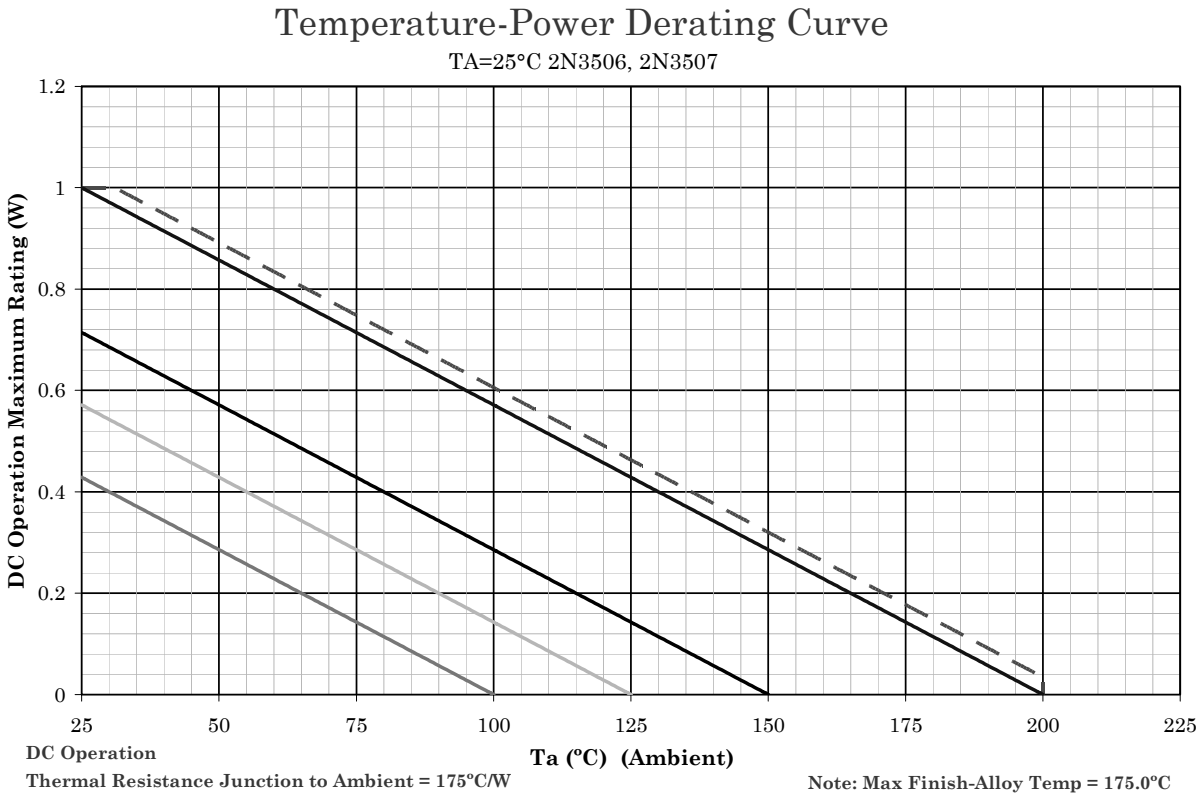
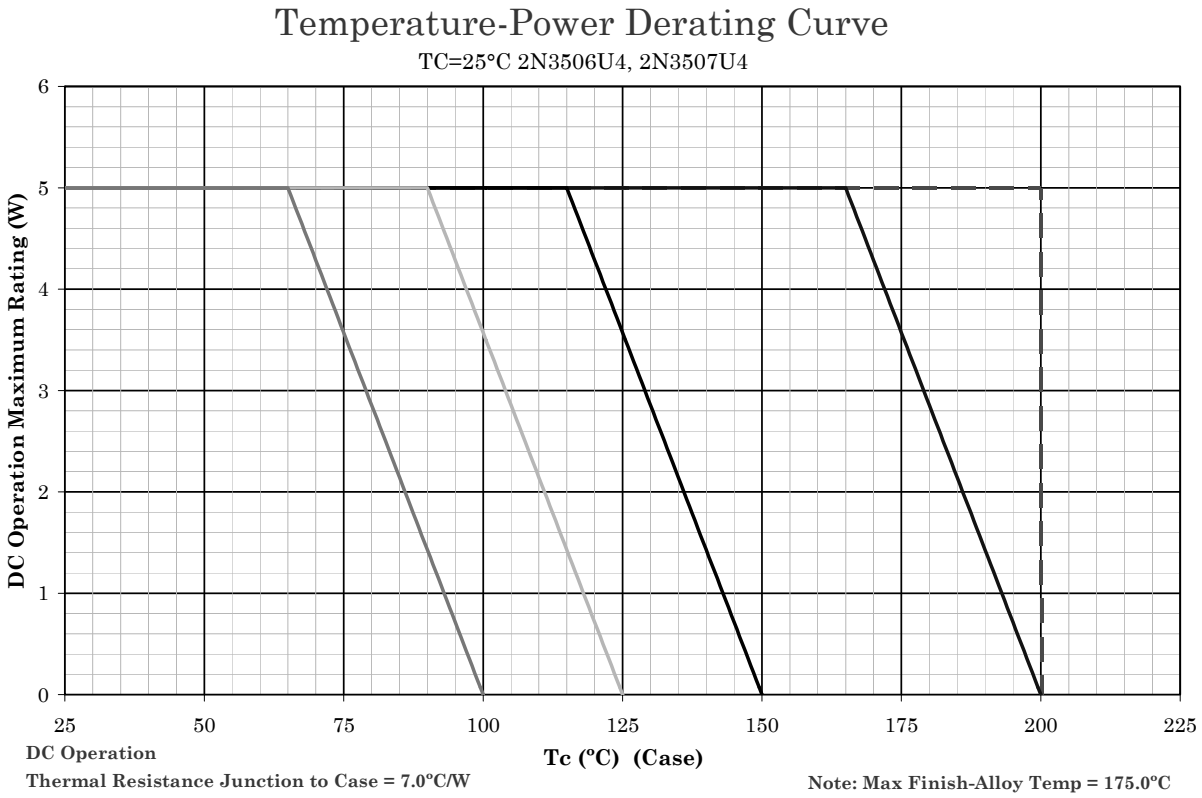


FIGURE 3. Temperature-power derating for 2N3506, 2N3506A, 2N3506L, 2N3506AL, 2N3507, 2N3507A, 2N3507L, and 2N3507AL.

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* FIGURE 4. Temperature-power derating for 2N3506U4 2N3507U4.

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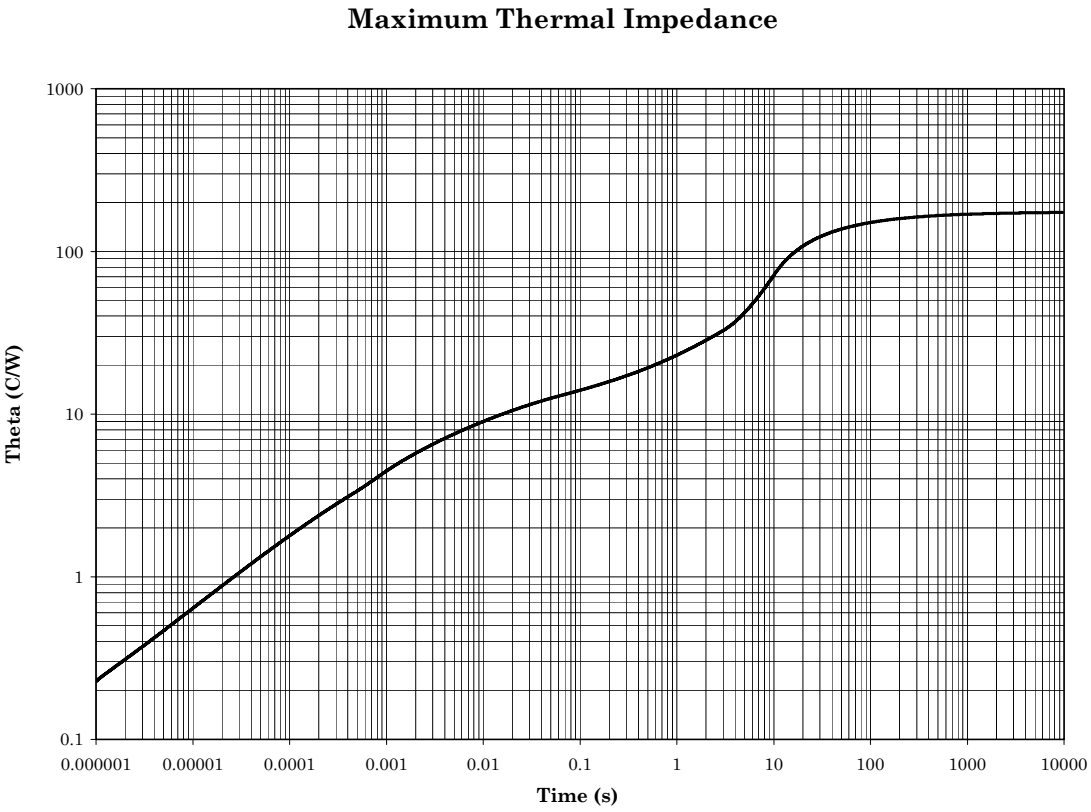


FIGURE 5. Thermal impedance graph ($R_{\theta JA}$) for 2N3506, 2N3506A, 2N3506L, 2N3506AL, 2N3507, 2N3507A, 2N3507L, and 2N3507AL.

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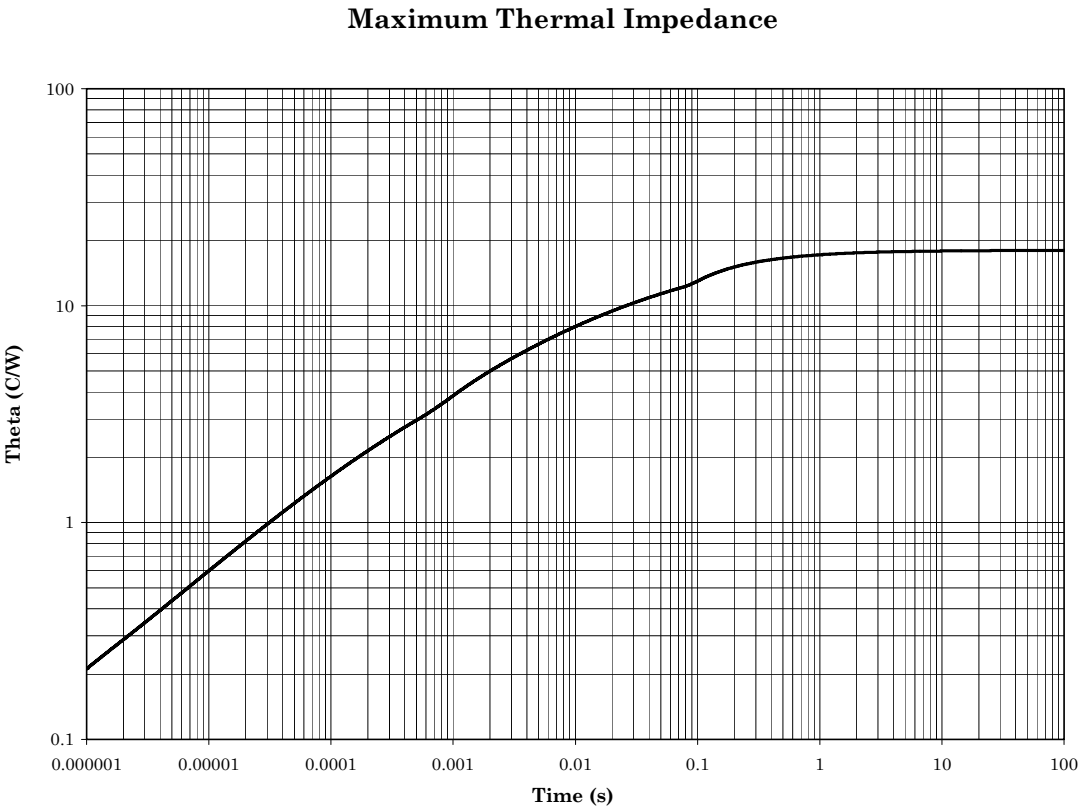
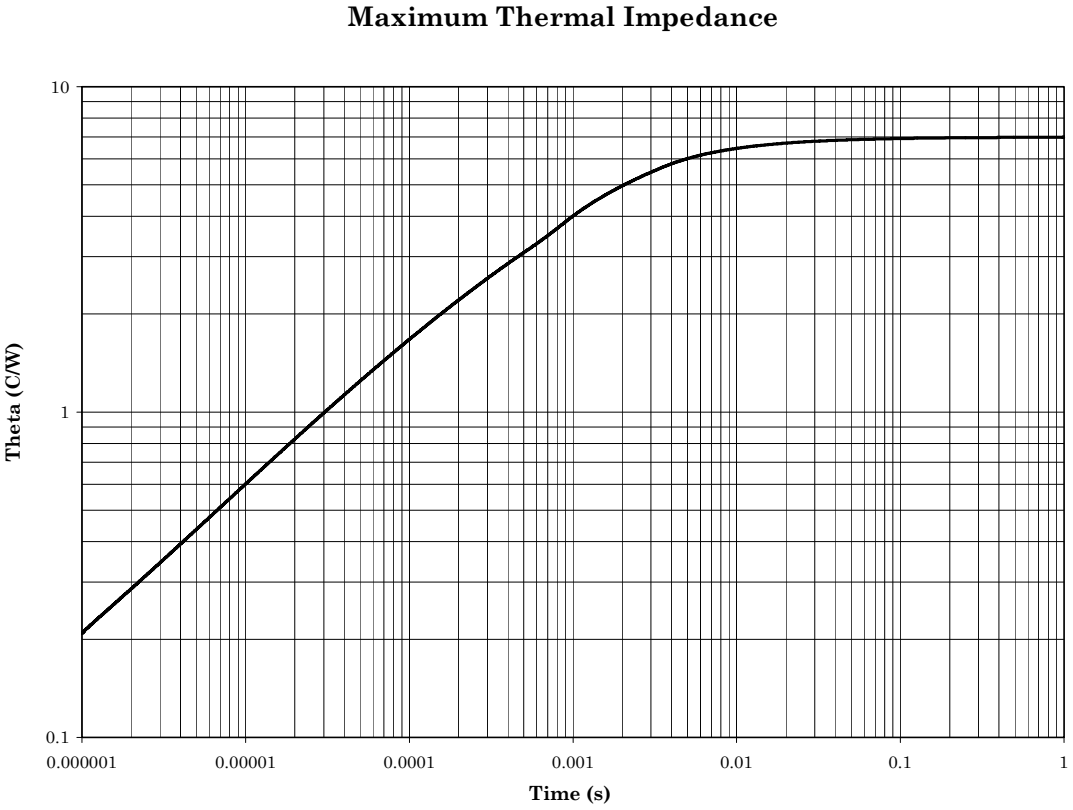


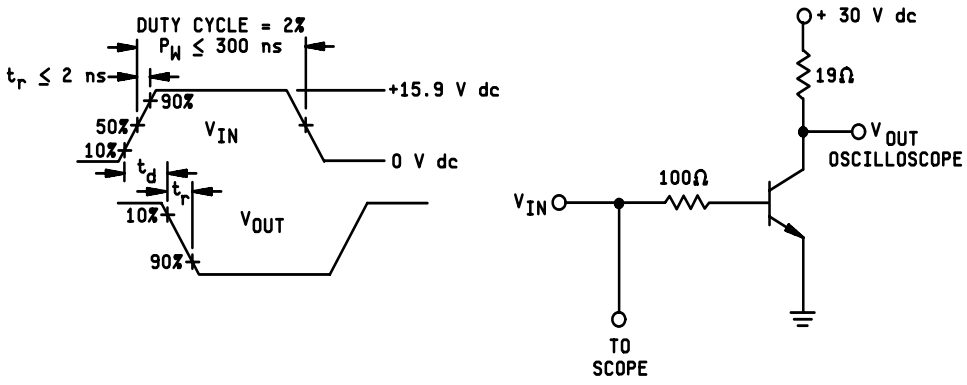
FIGURE 6. Thermal impedance graph ($R_{\theta JC}$) for 2N3506, 2N3506A, 2N3506L, 2N3506AL, 2N3507, 2N3507A, 2N3507L, and 2N3507AL.

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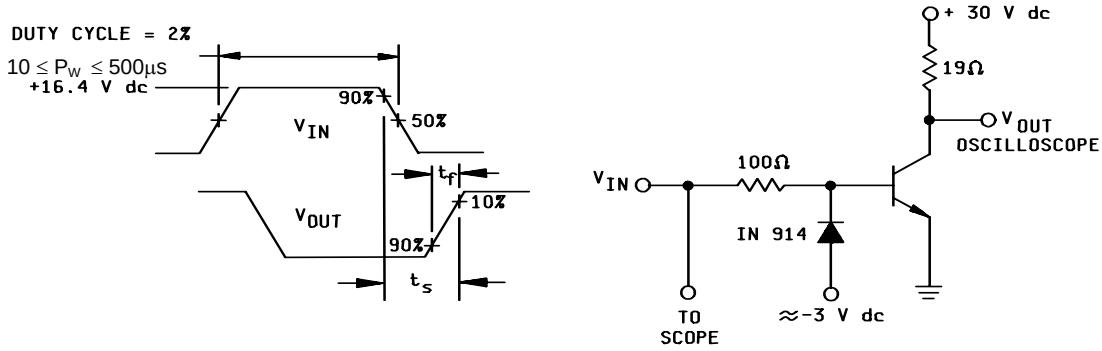


* FIGURE 7. Thermal impedance graph ($R_{\theta JC}$) for 2N3506U4 and 2N3507U4.

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* FIGURE 8. Equivalent circuit for measuring delay and rise times.



* FIGURE 9. Equivalent circuit for measuring storage and fall times.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Substitution information. The 2N3506A and 2N3507A may be substituted for the 2N3506 and 2N3507 (and L suffix versions) (respectively) at the discretion of the procuring activity."

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NASA - NA
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Review activities:
Army - AR, AV, MI
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Preparing activity:
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* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .

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